Note 10 ARIMA Models

1 ARIMA Models

Recall: An ARMA(p, q) model is

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} - \cdots - \phi_p X_{t-p} = C + Z_t - \theta_1 Z_{t-1} - \theta_2 Z_{t-2} - \cdots - \theta_q Z_{t-q},$$

Here $\{Z_t\}$ is white noise with mean zero and variance σ_Z^2. An ARMA(p, q) model in terms of the difference function is

$$\phi(B)X_t = C + \theta(B)Z_t.$$

For a time series $\{X_t\}$, if its dth order difference $\{\nabla^d X_t\}$ is an ARMA(p, q) processes, then $\{X_t\}$ is called an **Autoregressive Integrated Moving-Average model** with AR order p, trend-difference order d, and MA order q, or ARIMA(p, d, q).
In terms of difference function, an ARIMA(p, d, q) model is

$$\phi(B)(1 - B)^d X_t = C + \theta(B)Z_t.$$

Here, $\{Z_t\}$ is white noise with mean zero and variance σ_Z^2.

- For $d \neq 0$, ARIMA models are not stationary.

- If $\{X_t\}$ is an ARIMA(p, d, q) process, and $Y_t = \nabla^d X_t$, then $\{Y_t\}$ is an ARMA(p, q) process.

- Since ARIMA models are non-stationary (unless $d = 0$), we usually don't discuss the ACF and PACF of ARIMA models.

- The simplest ARIMA example is a random walk.

2 Parameter Redundancy

3 Causality

4 Invertibility

5 Building ARIMA Models

There are a few basic steps to fitting ARIMA models to time series data.

1. Plot the data.
2. If necessary, transform the data.
3. Identify the order of the model.

4. Parameter estimation.

5. Model selection and diagnostics.

6. Forecast.